



Basic Principles of Medicine II

Module: Nervous System & Behavioral Sciences

NB IMCQ 7

Sectional Anatomy of the CNS to
Lifespan Development (Age 0 – 2 years)



Behavioral Sciences

1) A 6-month-old boy is brought to the physician by his mother for a routine examination. He was born at 39 weeks' gestation via an uncomplicated spontaneous vaginal delivery. He is meeting all age-appropriate developmental milestones and is breastfeeding well. While sitting on his mother's lap, the boy accidentally strikes a toy attached to his stroller, causing it to light up and play music. He immediately strikes the toy again, appearing to enjoy the effect. He repeats this behavior several times during the encounter. Which of the following cognitive stages best explains this infant's repetitive behavior?

- A. Primary circular reaction
- B. Secondary circular reaction
- C. Tertiary circular reactions
- D. Preoperational
- E. Concrete operational
- F. Formal operational

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SOM.MK.III.BPM2.4.NB.3.BEHS.0005 Identify the major physical, cognitive, and social/emotional milestones of the first year

2) A 6-month-old girl is brought to the physician for a routine examination. She was born at term following an uncomplicated pregnancy and delivery. She sits on the table without support during the examination and tries to speak with the physician using repetitive consonant-vowel sounds. She reaches for the physician's stethoscope but is unable to grasp it or hold it in her hand. She cannot crawl or pull herself into a standing position. A nurse, whom she has not seen before, enters the room and tries to pick her up, causing her to cry and cling to her mother. When the physician plays "peek-a-boo" by hiding her toys, the child cries and does not look for the hidden toys. Which of the following developmental findings most likely requires further evaluation by the physician?

- A. Babbling
- B. Reaching for objects
- C. Cannot grasp objects
- D. Cannot crawl
- E. Crying around strangers
- F. Crying when toys are hidden

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SOM.MK.III.BPM2.4.NB.3.BEHS.0005

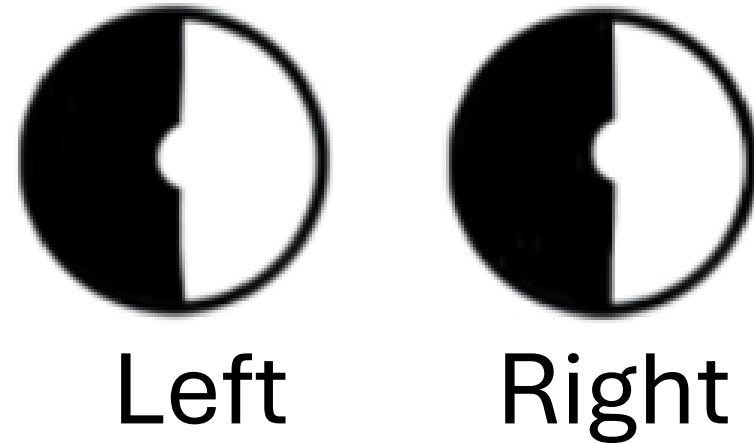
Identify the major physical, cognitive, and social/emotional milestones of the first year



Neuroscience

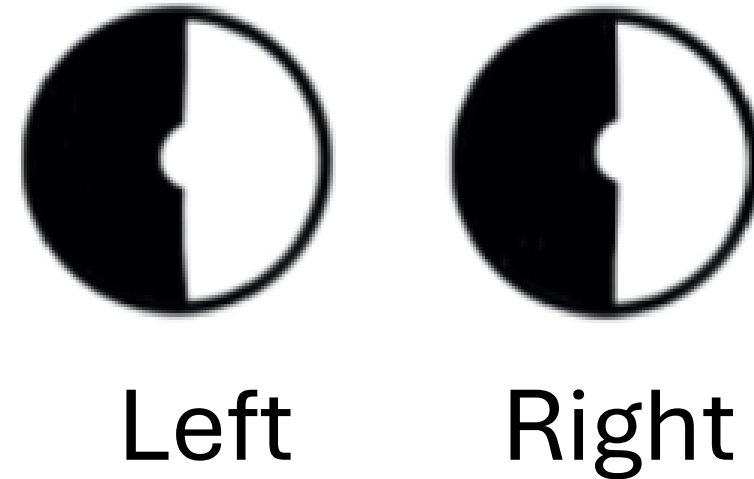
3. A 28-year-old woman comes to the physician because of a 2-day history of vision problems. Neurologic examination and clinical findings show visual field defects in the left and right eyes, as indicated in the figure below. Which of the following patterns of responses in pupillary light reflex testing is most likely in this patient?

- A. Light in left eye: Direct but not consensual responses
- B. Light in left eye: Consensual but not direct responses
- C. Light in left eye: Neither direct nor consensual responses
- D. Light in right eye: Direct but not consensual responses
- E. Light in right eye: Consensual but not direct responses
- F. Light in right eye: Neither direct nor consensual responses
- G. Light in right eye: Direct and consensual responses



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- G. Light in right eye: Direct and consensual responses



4. A 28-year-old man comes to the physician because of a 2-week history of double vision and headaches. During neutral gaze, his right eye is slightly elevated. Neurologic examination shows increased double vision when he attempts to gaze leftward and downward during the H-test. Which of the following structures is most likely lesioned in this patient?

- A. Left corticobulbar fibers
- B. Right corticobulbar fibers
- C. Left cranial nerve III
- D. Right cranial nerve III
- E. Left cranial nerve IV
- F. Right cranial nerve IV
- G. Left cranial nerve VI
- H. Right cranial nerve IV

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SOM.MK.I.BPM2.1.NB.4.NSCI.0235

Relate the clinical correlations of Trochlear Nerve Palsy, Oculomotor Nerve Palsy, Abducens Nerve Palsy, Diplopia, Internuclear Ophthalmoplegia, PPRF Lesion and One-and-a-half Syndrome to the underlying functional and anatomical principles

5. A 27-year-old unconscious man is examined by the physician at the site of a motor vehicle collision. Neurologic examination shows a dilated pupil on the right, responding sluggishly to light. A supratentorial hemorrhage with uncal herniation is suspected. Which of the following fibers of the autonomic nervous system are most likely compressed by the herniated structure in this patient?

- A. Left parasympathetic postganglionic
- B. Left parasympathetic preganglionic
- C. Left sympathetic postganglionic
- D. Left sympathetic preganglionic
- E. Right parasympathetic postganglionic
- F. Right parasympathetic preganglionic
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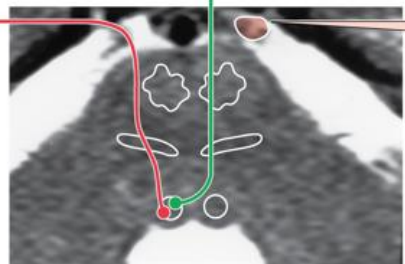
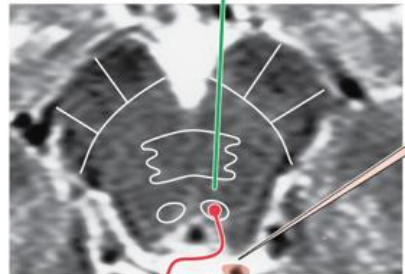
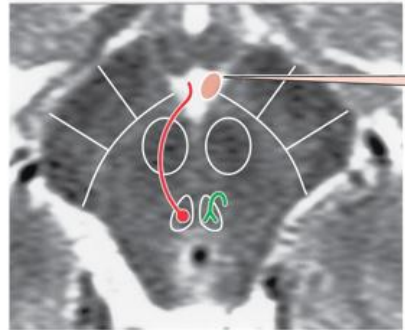
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- F. Right parasympathetic preganglionic
- G. Right sympathetic postganglionic
- H. Right sympathetic preganglionic

SOM.MK.I.BPM2.1.NB.1.NSCI.0274

Describe regulation of the autonomic nervous system by circuitry within the brain.

6. A 31-year-old woman comes to the physician because of a 2-day history of double vision, which gets worse when she tries to look to the right. When performing the H-test, during attempted gaze to the right, the right eye abducts, but the left eye stays in mid-position. The convergence test shows normal adduction of both eyes. Based on the images below, which of the following is the most likely location of the lesion?

- | | |
|----|---|
| A. | A |
| B. | B |
| C. | C |
| D. | D |
| E. | E |
| F. | F |
| G. | G |
| H. | H |

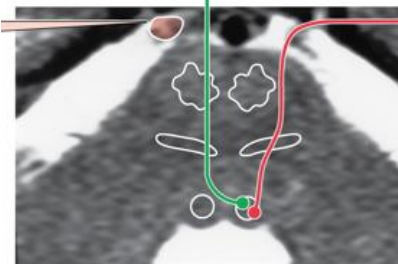
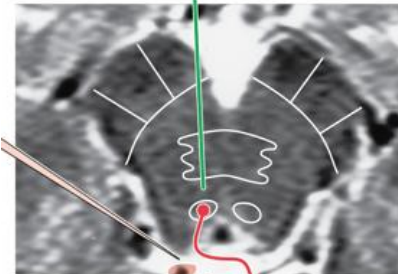
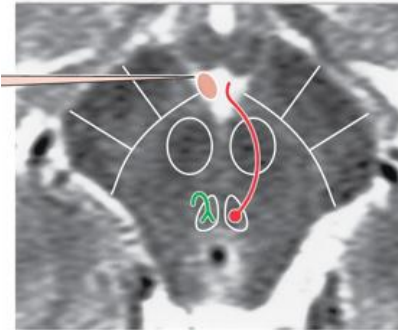


A - CN III root - B

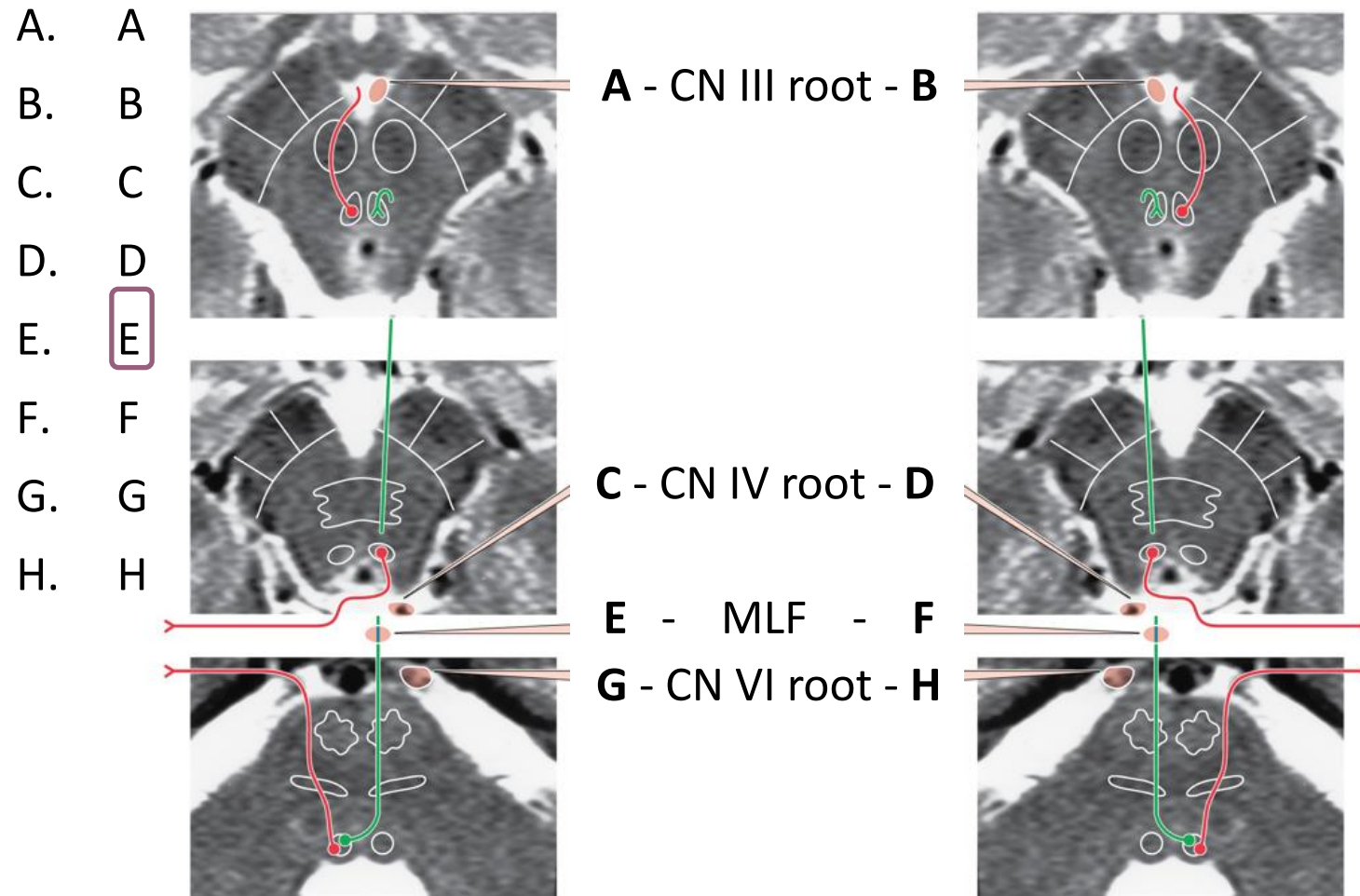
C - CN IV root - D

E - MLF - F

G - CN VI root - H



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Relate the clinical correlations of Trochlear Nerve Palsy, Oculomotor Nerve Palsy, Abducens Nerve Palsy, Diplopia, Internuclear Ophthalmoplegia, PPRF Lesion and One-and-a-half Syndrome to the underlying functional and anatomical principles

7. A 62-year-old man comes to the physician because of a 4-day history of headaches and visual problems. His vital signs are within normal limits. Imaging studies show a lesion in the right parietal lobe. Which of the following is the most likely visual deficit in this patient?

- A. Bitemporal hemianopia
- B. Left inferior quadrantanopia
- C. Left monocular blindness
- D. Left superior quadrantanopia
- E. Right inferior quadrantanopia
- F. Right monocular blindness
- G. Right superior quadrantanopia

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SOM.MK.I.BPM2.1.NB.4.NSCI.0228

Relate the clinical correlations of Visual Pathway Lesions, with and without Sparing of the Macula, and including Achromatopsia, alexia and prosopagnosia to the underlying functional and anatomical principles

8. A 61-year-old woman is brought to the physician because of a 2-hour history of weakness on the right side of the body. Neurologic examination shows hyperreflexia on the right and right extensor plantar response. Her left eye is adducted during primary gaze. Her tongue deviates to the right upon protrusion but is not atrophic. Which of the following is the most likely location of the lesion?

- A. Left midbrain
- B. Left medulla
- C. Left pons
- D. Right midbrain
- E. Right medulla
- F. Right pons

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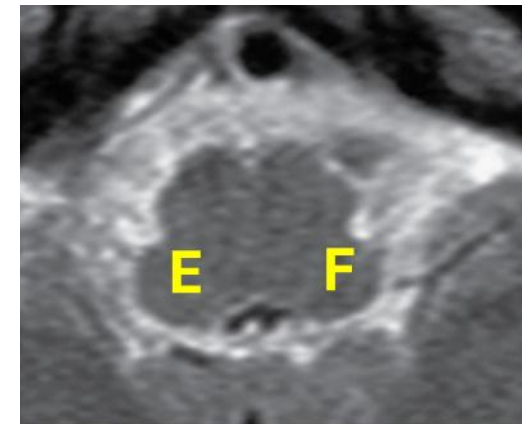
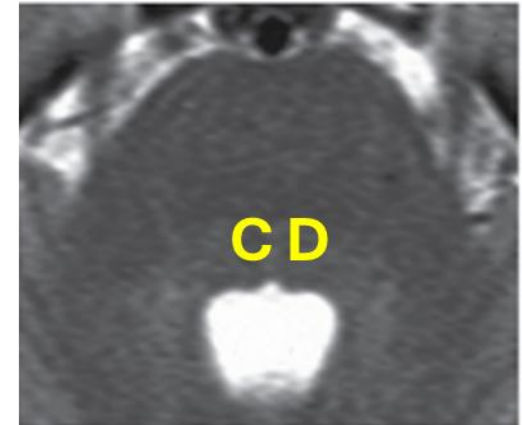
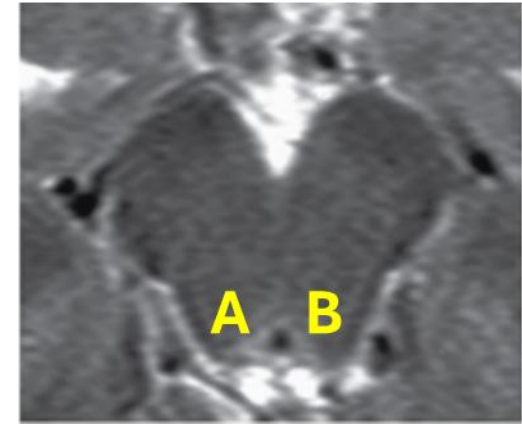
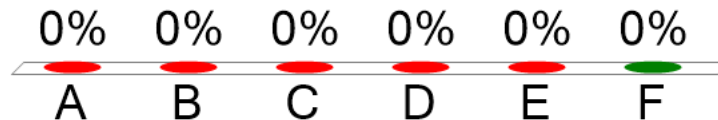
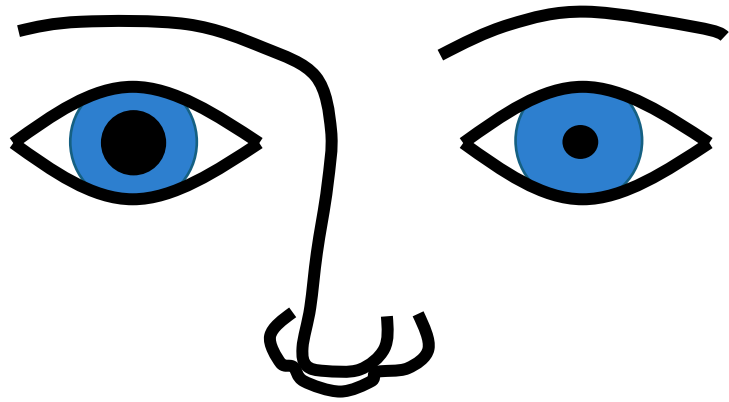
SOM.MK.I.BPM2.1.NB.4.NSCI.0534

Discuss general principles of lesion localization, based on functional assessment of nerves

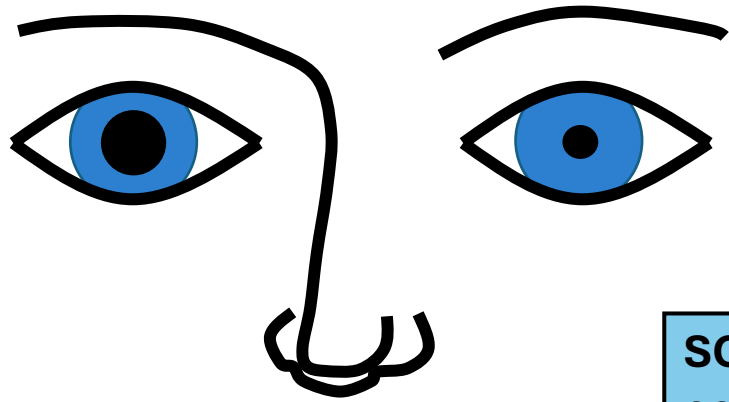
SOM.CS.II.BPM2.4.NB.4.NSCI.0538

Identify metencephalic lesions by assessment of cranial nerves

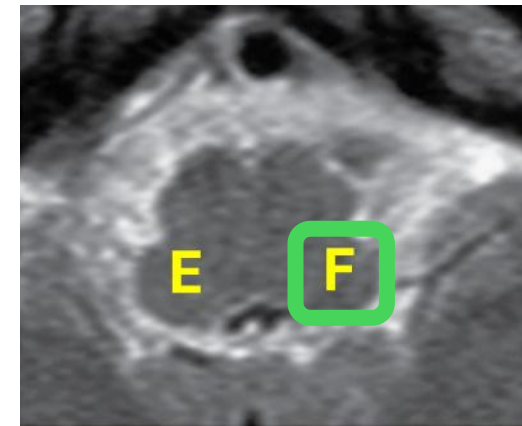
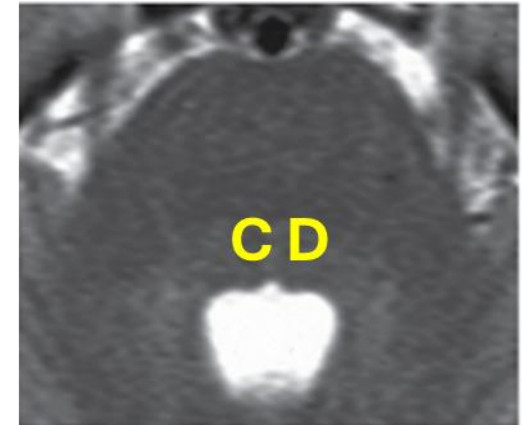
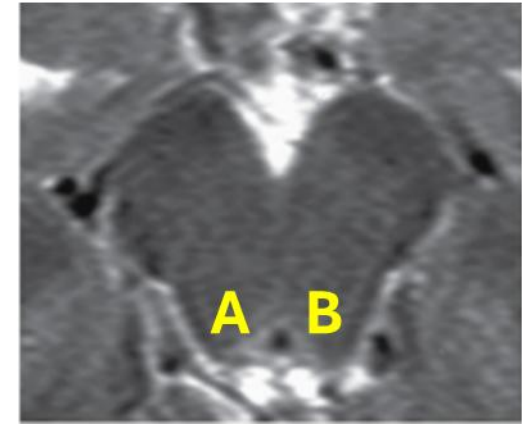
9. A 44-year-old woman comes to the physician because of a sudden onset of uncontrollable jerking eye movements. She feels nauseous and has vomited a few times. Her voice sounds hoarse, and her gait is unstable. Her left pupil is abnormally constricted, and her left eyelid is drooping slightly. Neurologic examination shows pathological nystagmus (see animation below). Which of the areas marked in the MRIs below is most likely damaged?



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SOM.MK.II.BPM2.1.NB.4.NSCI.0253
correlate Motion Sickness, Alcohol Intoxication, Antibiotics, and the Pathological Vestibular Nystagmus to the underlying functional and anatomical principles <Required Reading Assignment>

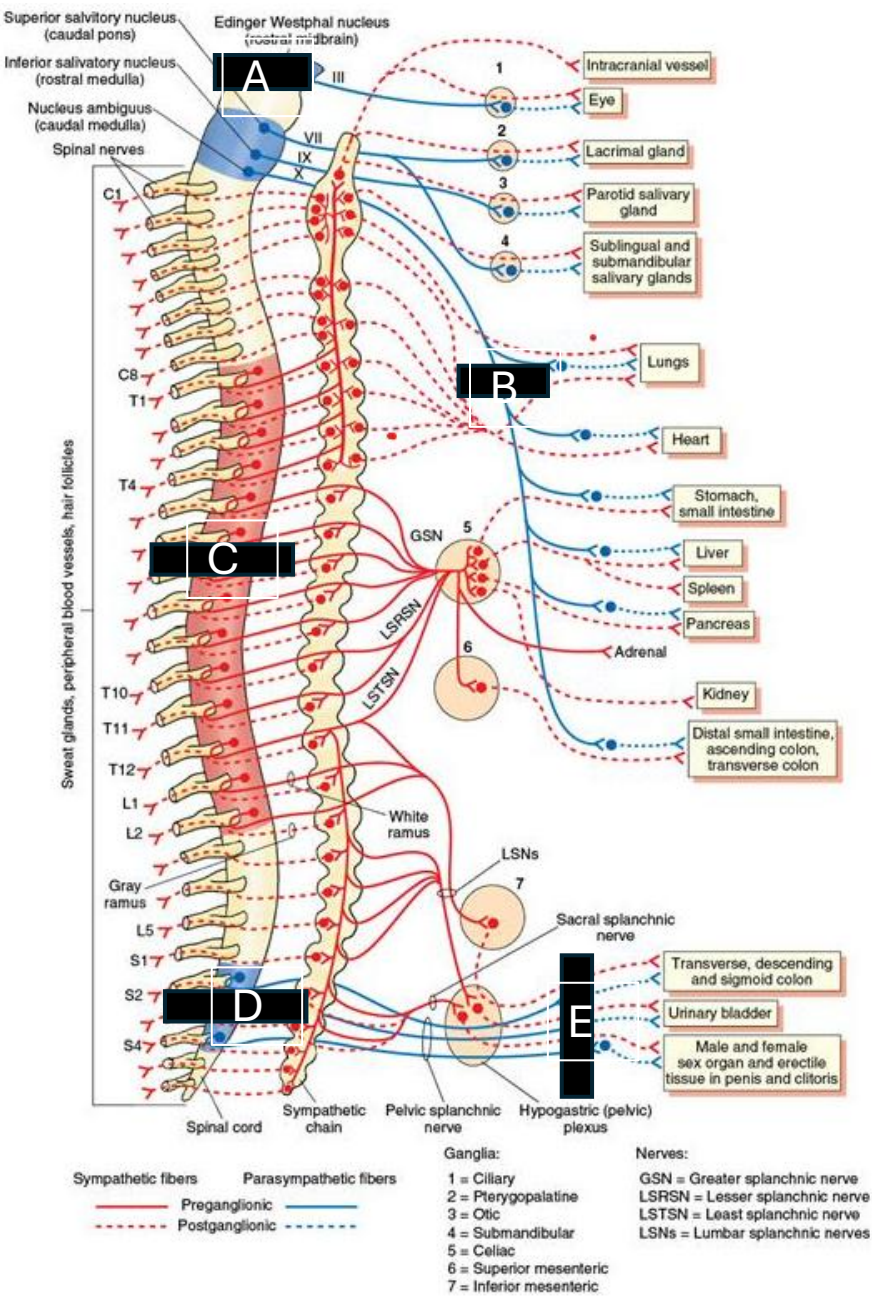


10. A 65-year-old man is evaluated at the hospital three-months after a gun-shot injury. He is wheelchair bound and emptying his bladder spontaneously. His vitals are within the normal range. He complains of frequent spontaneous urination. His pupillary light reflex is intact bilaterally. Kneejerk reflex is exaggerated, and plantar response is extensor bilaterally. Imaging (attached) shows elongated small capacity urinary bladder with an irregular trabeculated wall. Which of the following areas marked in the schematic diagram best describes the location of the lesion?



Case courtesy of Ian Bickle, Radiopaedia.org, rID: 28323

- A. A
- B. B
- C. C
- D. D
- E. E

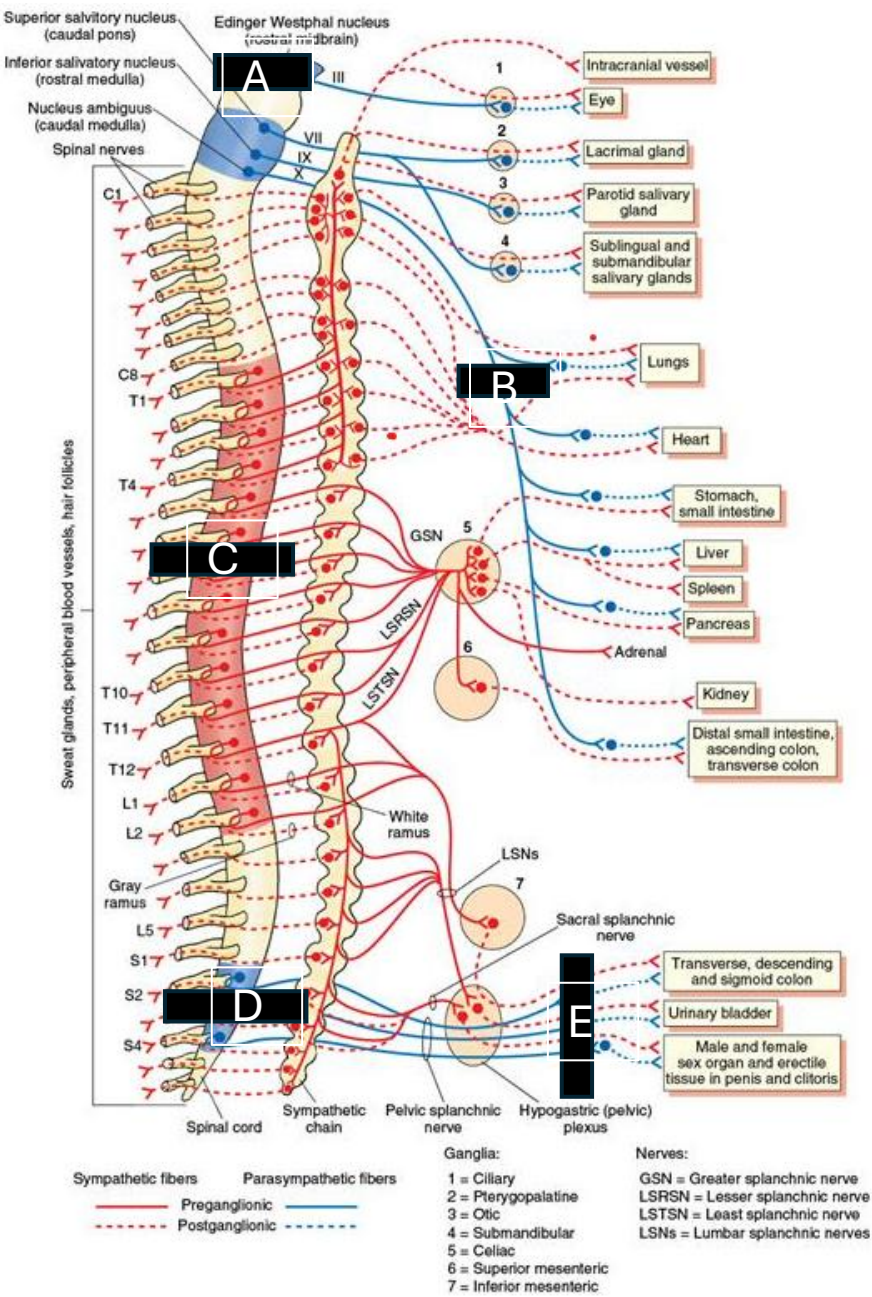


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- A. A
- B. B
- C. C
- D. D
- E. E



SOM.MK.I.BPM2.4.NB.2.NSCI.0280
Contrast automatic and atonic bladder

11. A 45-year-old woman comes to the physician because of a 2-month history of hearing difficulty in the right ear. Neurologic examination shows Weber's test lateralizing to the right, bone conduction greater than air conduction on the right, and air conduction greater than bone conduction on the left. Which of the following is the most likely location of the lesion?

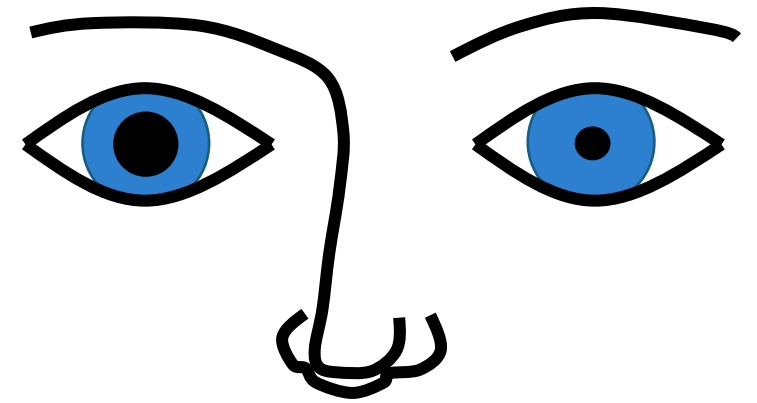
- A. Cochlear nerve
- B. Inferior colliculus
- C. Lateral lemniscus
- D. Transverse temporal gyrus
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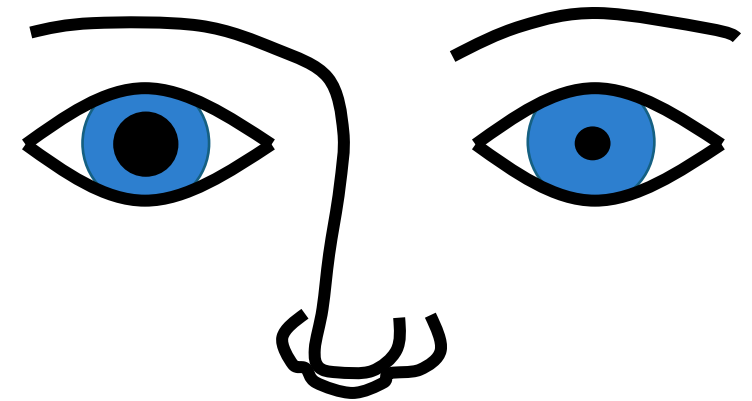
12. A 44-year-old woman comes to the physician because of a sudden onset of uncontrollable jerking eye movements. She feels nauseous and has vomited a few times. Her voice sounds hoarse, and her gait is unstable. Her left pupil is abnormally constricted, and her left eyelid is drooping slightly. Neurologic examination shows pathological nystagmus (see animation below). Which of the following additional symptoms are most likely expected in this patient?

- A. Left UE and LE weakness
- B. Right UE and LE weakness
- C. Left tongue deviation upon protrusion
- D. Right tongue deviation upon protrusion
- E. Left UE and LE loss of proprioception
- F. Right UE and LE loss of proprioception
- G. Left loss of facial pain
- H. Right loss of facial pain



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SOM.MK.I.BPM2.4.NB.4.NSCI.0122

Use the knowledge of pain and temperature pathways in a clinical scenario to localize a lesion <Required Reading Assignment>

13. A 72-year-old woman comes to the physician because of sudden-onset clumsiness. She exhibits right-sided dysdiadochokinesia, dysmetria and intention tremor. Speech is hesitant and articulation is poor. Which of the cerebellar sites indicated in the image most likely overlaps with the site of CNS injury in this patient?

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4



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SOM.MK.I.BPM2.4.NB.2.NSCI.0202

Correlate cerebellar lesion sites and clinical manifestations

14. A 71-year-old man comes to the physician because of a 7-month history of progressively worsening hearing loss. Tuning fork tests show Weber's test lateralizes to the right; Rinnes's test AC > BC bilaterally. Muscles of facial expression and mastication are weak on the left half of the face, and reduced touch and nociceptive sense is present in the same distribution. Which of the following is the most likely cause of the findings in this patient?

- A. Bell's palsy
- B. AICA occlusion
- C. Presbycusis
- D. Temporal lobe tumor
- E. Vestibular schwannoma

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SOM.MK.I.BPM2.1.NB.4.NSCI.0534

Discuss general principles of lesion localization, based on functional assessment of nerves

SOM.CS.II.BPM2.4.NB.4.NSCI.0538

Identify metencephalic lesions by assessment of cranial nerves

SOM.CS.I.BPM2.1.NB.4.NSCI.0244 Describe clinical assessment of auditory function